

Chapter 8 Fluid mechanics.

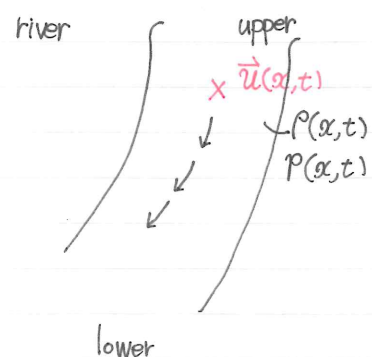
Chapter 8. Fluid mechanics

We study fluid mechanics in $\mathbb{R}^n$

Denote $\vec{u} = \vec{u}(x, t)$: ~~the~~ velocity vec. of fluid.

$\rho = \rho(x, t)$: density of fluid.

$P = P(x, t)$: pressure of fluid.



<coordinates>

- Eulerian coordinate: look at the fluid at some fixed pt.
- Lagrangian coordinate: look at the fluid ~~at~~ of the fluid (moving)

Today we use Eulerian coordinate

$\Rightarrow$ For any pt $x \in \mathbb{R}^n$ and any time $t > 0$. study $\vec{u} = \vec{u}(x, t)$

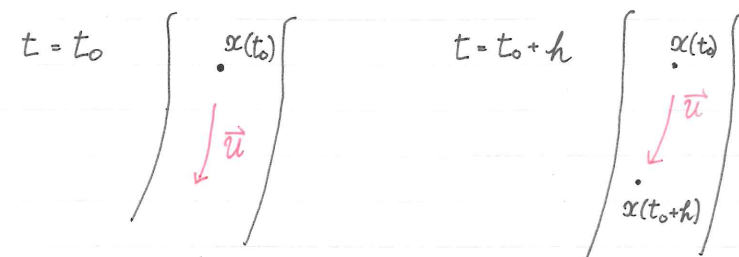
c.f. For Lagrangian coordinate.

study $\vec{u} = \vec{u}(\underset{\alpha(x, t)}{*}(x, t), t)$, $\rho = \rho(\alpha(x, t), t)$, $P = P(\alpha(x, t), t)$.

for some starting pt $X \in \mathbb{R}^n$.

<Material derivative>

For example we consider the density



difference of the density

$$\rightarrow \rho(x + \vec{u}h, t_0 + h) - \rho(x, t_0).$$

rate of the density

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{\rho(x + \vec{u}h, t_0 + h) - \rho(x, t_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\rho(x + \vec{u}h, t_0 + h) - \rho(\vec{u}h, t_0)}{h} + \lim_{h \rightarrow 0} \frac{\rho(\vec{u}h, t_0) - \rho(x, t_0)}{h} \\ &= \frac{\partial \rho}{\partial t}(x, t) + \vec{u}(x, t) \cdot \nabla \rho(x, t). \end{aligned}$$

Here we call

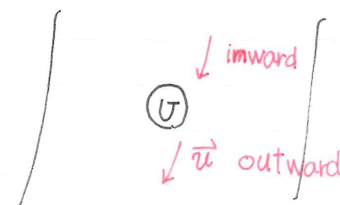
$$D = \frac{\partial}{\partial t} + \vec{u} \cdot \nabla$$

material derivative.

<conservation of mass>

For a set $U \subset \mathbb{R}^n$, mass on U can be denoted as

$$\int_U \rho(x, t) dx$$



At a time t the mass on U is going away of.

$$\int_{\partial U} \underset{\text{density}}{\rho(\alpha, t)} \underset{\text{velocity}}{\vec{u}(\alpha, t)} \cdot \underset{\text{outer unit normal}}{\vec{v}(\alpha, t)} d\sigma.$$

Conservation of mass

$$\frac{d}{dt} \int_U \rho(\alpha, t) d\alpha$$

$$= - \int_{\partial U} \rho(\alpha, t) \vec{u}(\alpha, t) \cdot \vec{v}(\alpha, t) d\sigma$$

$$\Leftrightarrow \int_{\partial U} \frac{\partial \rho}{\partial t}(\alpha, t) d\alpha$$

$$+ \int_{\partial U} \rho(\alpha, t) \vec{u}(\alpha, t) \cdot \vec{v}(\alpha, t) d\sigma$$

$$\stackrel{\text{div thm}}{=} \int_{\partial U} \frac{\partial \rho}{\partial t}(\alpha, t) d\alpha + \int_U \operatorname{div}(\rho(\alpha, t) \vec{u}(\alpha, t)) d\alpha = 0.$$

$$\therefore \int_U \left(\rho_t(\alpha, t) + \operatorname{div}(\rho(\alpha, t) \vec{u}(\alpha, t)) \right) d\alpha = 0$$

$$\Rightarrow \rho_t(\alpha, t) + \operatorname{div}(\rho(\alpha, t) \vec{u}(\alpha, t)) = 0 \quad (8.2)$$

(8.2) is called conservation of mass
or continuity equation.

Remark: if $\rho \equiv 1$, then $\rho_t(\alpha, t) = 0$.

$$\text{Thus } (8.2) \Rightarrow \operatorname{div}(\vec{u}(\alpha, t)) = 0$$

2nd. order eq. of NS.

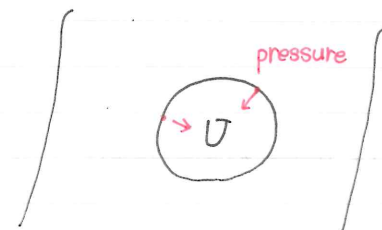
$$\text{c.f. } \vec{u} = (u, v, w)$$

$$\operatorname{div} \vec{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$$

< conservation of momentum >

For a set of $U \subset \mathbb{R}^n$ the total pressure on U is

$$\int_{\partial U} P(\alpha, t) \cdot (-\vec{v}(\alpha, t)) d\sigma$$



By d'Alembert' law

$$(\text{pressure}) \times (\text{outer other force}) - (\text{mass}) \times (\text{acceleration}) = 0.$$

Let $\vec{K}(\alpha, t)$ be outer force

Then by d'Alembert' law

$$\int_{\partial U} P(\alpha, t) \cdot (-\vec{v}(\alpha, t)) d\sigma + \int_U \vec{K}(\alpha, t) d\alpha$$

$$- \int_U \rho(\alpha, t) \frac{D\vec{u}}{Dt}(\alpha, t) d\alpha = 0.$$

$$\left(\frac{D}{Dt} = \frac{\partial}{\partial t} + \vec{u} \cdot \nabla \right).$$

By div thm

$$\int_{\partial U} P(\alpha, t) \cdot (-\vec{v}(\alpha, t)) d\sigma = - \int_U \nabla P(\alpha, t) d\alpha$$

$$\therefore \int_U \left(-\nabla P(\alpha, t) + \vec{K}(\alpha, t) - \rho(\alpha, t) \frac{D\vec{u}}{Dt}(\alpha, t) \right) d\alpha = 0.$$

for ~~any~~ arbitrary $U \subset \mathbb{R}^n$

$$\Rightarrow \int_U \rho(\alpha, t) \frac{D\vec{u}}{Dt}(\alpha, t) + \nabla p(\alpha, t) = \vec{K}(\alpha, t) \quad (8.5)$$

(8.5) is called "law of conservation of momentum."

Remark: $\rho \equiv 1$, then

$$(8.5) \Rightarrow \frac{D\vec{u}}{Dt}(\alpha, t) + \nabla p(\alpha, t) = \vec{K}(\alpha, t)$$

<Euler eq>

$$\left\{ \begin{array}{l} \frac{\partial \vec{u}}{\partial t}(\alpha, t) + \vec{u}(\alpha, t) \cdot \nabla \vec{u}(\alpha, t) + \nabla p(\alpha, t) = \vec{K}(\alpha, t) \\ \text{div}(\vec{u}(\alpha, t)) = 0 \end{array} \right. \quad \begin{array}{l} \text{momentum} \\ \text{mass} \end{array}$$

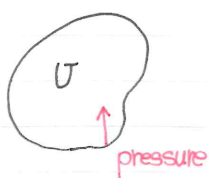
(E)

(E) is called Euler eq.

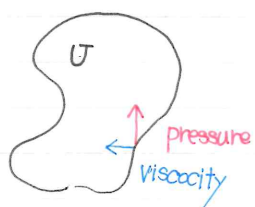
<NS eq>

For the Euler eq. we consider pressure as a force but not only pressure. but also other effect (viscosity) should be considered.

• Euler eq



• NS



<Idea / strategy>

Change $p(\alpha, t)(-\vec{\nabla}(\alpha))$ to stress tensor $\vec{P}(\alpha, t)$

pressure + viscosity + ...

If $\vec{P}(\alpha, t) = P(\alpha, t)(-\vec{\nabla}(\alpha))$

stress tensor (matrix)

then

$$\int_{\partial U} P(\alpha, t)(-\vec{\nabla}(\alpha)) d\sigma$$

$$= - \left(\int_U \sum_{j=1}^n \frac{d}{d\alpha_j} p^{ij}(\alpha, t) d\alpha \right)_{i=1,2,\dots,n}$$

where $P^{ij} = (p_{ij})$

Let

$$P = pI - \lambda \operatorname{div} \vec{u} I - \mu \left(\frac{\partial u^i}{\partial \alpha_j} + \frac{\partial u^j}{\partial \alpha_i} \right)$$

then momentum eq turns into

$$\rho \frac{D\vec{u}}{Dt}(\alpha, t) - \mu \Delta \vec{u} - (\lambda + \mu) \nabla \operatorname{div} \vec{u} + \nabla p = \vec{K}$$

→ see note for the detail.

<NS eq> $\rho \equiv 1$.

$$\left\{ \begin{array}{l} \frac{\partial \vec{u}}{\partial t}(\alpha, t) + \underbrace{\vec{u}(\alpha, t) \cdot \nabla \vec{u}(\alpha, t)}_{\text{non-lin}} - \mu \Delta \vec{u}(\alpha, t) + \nabla p(\alpha, t) = \vec{K}(\alpha, t) \\ \operatorname{div}(\vec{u}(\alpha, t)) = 0 \end{array} \right.$$

(NS)

(NS) is called Navier-Stokes eq.